Mathematical Analysis 2

Multivariable derivatives 2

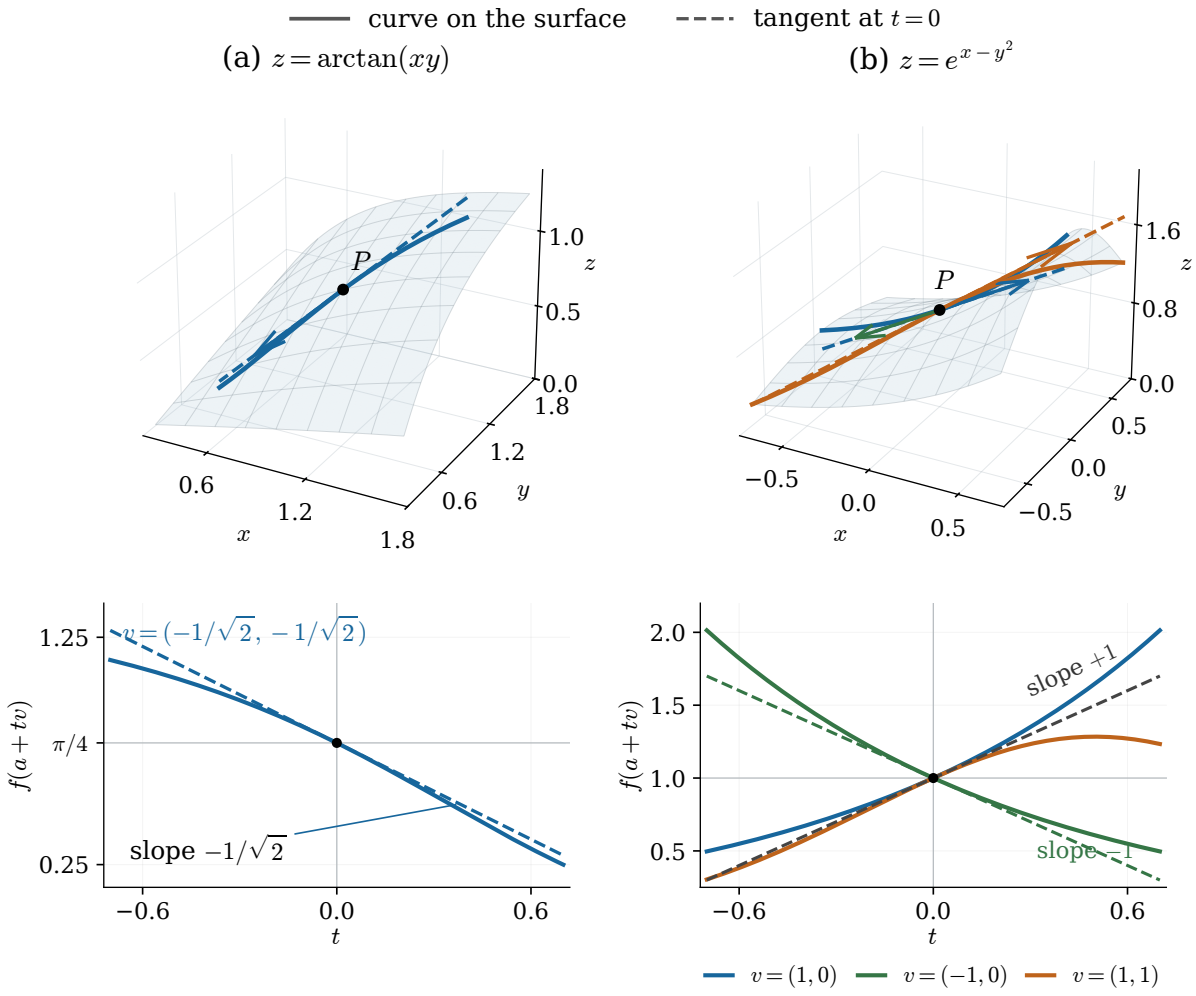
Peter Zeman

1. Verify directly from the definition that the linear function $L(h_1, h_2) = 2(h_1 + h_2)$ is the total differential of $f(x, y) = x^2 + y^2$ at $(1, 1)$.

2. Compute the directional derivatives of the following functions in the given directions at the specified points.

   (a) $f(x, y) = \arctan(xy)$ in the direction $v = (-1/\sqrt{2}, -1/\sqrt{2})$ at $a = (1, 1)$.

   (b) $f(x, y) = e^{x-y^2}$ in the directions $(1, 0)$, $(-1, 0)$, and $(1, 1)$ at $(0, 0)$.



The upper plots show the surface paths $(a + tv, f(a + tv))$ through $P = (a, f(a))$; arrows indicate increasing t. The lower plots show the function value $f(a + tv)$ against t. The slope of the dashed tangent at $t = 0$ is $D_v f(a)$. Slopes are measured per unit t, with each direction used as given.

3. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be given by

$$f(x,y) = \begin{cases} 1, & x = 0 \text{ or } y = 0, \\ 0, & x \neq 0 \text{ and } y \neq 0. \end{cases}$$

   (a) Determine where f is continuous.

   (b) Compute the partial derivatives at points off the coordinate axes, at points on the axes other than $(0,0)$, and at $(0,0)$.

   (c) Does f have a total differential at $(0,0)$?

   (d) Compute the directional derivative in the direction $(1,1)$ at points off the coordinate axes and at $(0,0)$.

4. Consider the function

$$f(x,y) = \log(\sqrt{y+1} - x).$$

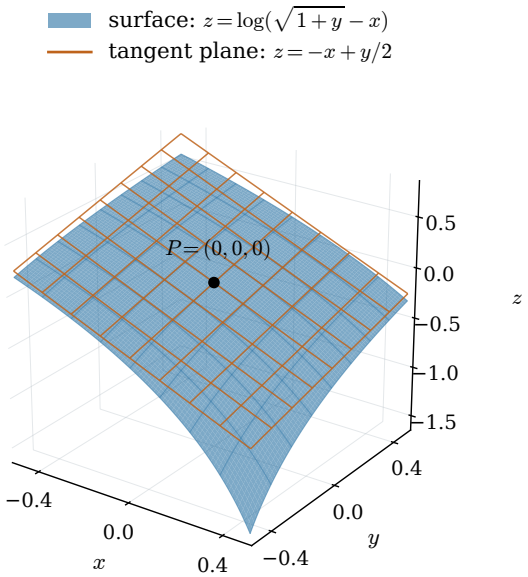
   (a) Find and sketch its domain D. Is f continuous?

   (b) Compute $\nabla f(x,y)$ wherever it exists, and determine $\nabla f(0,0)$.

   (c) Does f have a total differential at $(0,0)$? If so, find it.

   (d) Find the equation of the tangent plane at $(0,0)$.

   (e) Use linear approximation to estimate $f(-0.04, 0.02)$.



The blue surface and the orange tangent plane meet at $P = (0,0,0)$ with the same slope in every direction.

5. Determine whether

$$f(x,y) = \frac{x+y}{x^2+y^2} \ln(1+xy)$$

can be extended to a neighbourhood of the origin so that it has a total differential at the origin.

6. Find the directional derivatives of

$$f(x,y,z) = \sqrt[3]{x^3+y^3+z^3}$$

at $(0,0,0)$ in a direction $h = (h_1, h_2, h_3)$. Decide whether f has a total differential at this point.

7. Compute the total differentials of the following functions. How can they be used to approximate the functions near a point $a \in \mathbb{R}^2$ or $a \in \mathbb{R}^3$, respectively?

   (a) $f(x,y) = e^{xy}$.

   (b) $f(x,y,z) = xy + yz + xz$.

8. Show that, for small x and y,

   (a) $(1+x)^m(1+y)^n \approx 1 + mx + ny$;

   (b) $\ln(1+x)\ln(1+y) \approx xy$.

9. Find the equation of the tangent plane to the graph of $f(x,y) = \arctan(x/y)$ at $(1,1,f(1,1))$.

10. The following function is known to be continuous:

$$f(x,y) = \begin{cases} (x^2+y^2)\sin\left(\dfrac{1}{x^2+y^2}\right), & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

    Compute its partial derivatives wherever they exist. Are they continuous at $(0,0)$? Find a candidate for the total differential at this point and check whether it is indeed the total differential.

Additional practice

11. Consider the function

$$f(x,y) = 4\sqrt{1 - \frac{y}{x+1}}.$$

    (a) Find and sketch the domain of f.

    (b) Compute the partial derivatives wherever they exist.

    (c) Show that f is differentiable at $(0,-3)$ and find its total differential there.

    (d) Find the equation of the tangent plane to the graph of f at $(0,-3,8)$.

    (e) Estimate $f(-0.03,-3.01)$.

12. Find the total differential of $f(x,y) = x/y$ wherever it exists, and justify that it is indeed the total differential.

13. Determine whether the function

$$f(x,y) = \frac{x^3 y}{x^2+y^2}$$

    can be extended so that it has a total differential at every point of $\mathbb{R}^2$. Find the total differential wherever it exists.

14. Let $f(x,y) = -x^2 - y^2 + 2x + 4y - 4$. Find a tangent plane T to its graph that is perpendicular to the line $\{(t,t,t) \in \mathbb{R}^3 : t \in \mathbb{R}\}$. At which point does T intersect the line $\{(0,0,t) \in \mathbb{R}^3 : t \in \mathbb{R}\}$?